

The screenshot shows a SQL query execution interface. At the top, there are tabs for 'Query 4' and 'Query 5'. The active query is 'SELECT * FROM poc_db."orders_iceberg\$snapshots";'. Below the query editor, there are buttons for 'Run again', 'Explain', 'Cancel', 'Clear', and 'Create'. A status bar indicates 'SQL Ln 1, Col 49'. Below the query editor, there are tabs for 'Query results' and 'Query stats'. The 'Query results' tab is active, showing a green bar with 'Completed' status. Below this, there are statistics: 'Time in queue: 100 ms', 'Run time: 587 ms', and 'Data scanned: 0.59 KB'. There are also buttons for 'Copy' and 'Download results CSV'. Below the statistics, there is a search bar and a table of results. The table has columns: '#', 'committed_at', 'snapshot_id', 'parent_id', 'operation', and 'manifest_list'. The first row of data is: 1, 2026-08-01 09:07:04.968 UTC, 952695883030641274, append, s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.

Iceberg Table Cross-Platform POV

AWS Glue Catalog as Source of Truth - Accessed via Snowflake, Databricks, and Athena - Proof of Value Documentation

1. Objective

This is an internal exploration, not a specific client requirement. The goal is to understand, hands-on, how a single Apache Iceberg table - created and cataloged once in AWS Glue - can be accessed consistently from multiple analytics platforms, and whether changes made on one platform stay in sync when read from the others.

AWS Glue Data Catalog is used as the single shared catalog (source of truth). Three platforms will connect to it and read/write the same underlying Iceberg table:

- - Snowflake (via a Glue catalog integration)
- - Databricks (via Glue as external metastore/catalog)
- - Amazon Athena (native Glue integration)

What we want to learn:

- - Can all three platforms read the same Iceberg table registered in Glue without any data duplication or separate load step?
- - When an INSERT is run from one platform, how quickly and reliably does it appear when queried from the other two?
- - Same question for UPDATE and DELETE operations - do all platforms see a consistent row-level state after the change, or is there staleness/conflict?
- - Which platforms can only read vs. which can also write to the Glue-cataloged table?
- - Where does synchronization break down, if anywhere - metadata refresh delays, caching, permissions, or format-level limitations?

Scope of this POV:

- - Storage: Amazon S3 (single region) holding the Iceberg data/metadata files
- - Catalog: AWS Glue Data Catalog (single shared catalog for all three platforms)
- - Platforms under test: Snowflake, Databricks, Amazon Athena
- - Sample dataset only (not production data or volumes)



- Internal exploration - findings will inform whether/how this pattern gets proposed to a client later

Success criteria (what "done" looks like for this POV):

- A single Iceberg table, cataloged once in Glue, is queried successfully from Snowflake, Databricks, and Athena
- Insert, update, and delete operations performed on one platform are confirmed (with evidence/screenshots) as visible on the other two
- Any sync delay, conflict, or inconsistency observed is documented with the exact steps to reproduce it
- A clear internal recommendation on viability, plus any platform-specific limitations found (e.g. read-only vs. read-write support)

2. Architecture Overview

AWS S3 — Storage layer
 AWS Glue — Catalog
 Data Platform — Snowflake, Databricks and Athena

3. Storage Setup (Amazon S3)

Step 3.1: Create S3 Bucket

Bucket Name	ta-icebreg-poc
Region	eu-north-1
Versioning	Enabled
Encryption	SSE-S3
Block Public Access	True(All 4 settings)

Screenshot Reference:

Query 4
Query 5

1
SELECT * FROM poc_db."orders_iceberg\$snapshots"

SQL
Ln 1, Col 49

Run again
Explain
Cancel
Clear
Create

Reuse query results
up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 100 ms
Run time: 587 ms
Data scanned: 0.59 KB

Results (1)
Copy
Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-manifest-1

ta-icebreg-poc
Info

Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points

Bucket overview

AWS Region

Europe (Stockholm) eu-north-1

Amazon Resource Name (ARN)

arn:aws:s3:::ta-icebreg-poc

Creation date

July 30, 2026, 09:28:44 (UTC+05:30)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Enabled

Step 3.2: Define Base Iceberg Path

Base Path (s3://...)	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/
Warehouse Folder	warehouse/
Database Namespace	poc_db

Step 3.3: Create IAM Role & Policy

Role Name	iceberg-role
Permissions Policy Summary	Glue(Read, Write) and S3(List,Read, Write)
Trust Policy Summary	Added Snowflake IAM and Databrick IAM

Query 4 : X

Query 5 : X

(+)

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL

Ln 1, Col 49

Run again

▼

Explain

Cancel

Clear

Create

▼

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV

Search rows

#

committed_at

snapshot_id

parent_id

operation

manifest_list

1

2026-08-01 09:07:04.968 UTC

952695883030641274

append

s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Screenshot Reference :

icebreg_ta_policy

Info

Edit

Delete

Policy details

Type

Customer managed

Creation time

July 30, 2026, 09:48 (UTC+05:30)

Edited time

July 30, 2026, 09:48 (UTC+05:30)

ARN

arn:aws:iam::381492064215:policy/icebreg_ta_policy

Permissions

Entities attached

Tags

Policy versions (1)

Last Accessed

Permissions defined in this policy

Info

Edit

Summary

JSON

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Search

Allow (2 of 472 services)

Show remaining 470 services

Service

Access level

Resource

Request condition

Glue

Limited: Read, Write

All resources

None

S3

Limited: List, Read, Write

Multiple

s3:prefix| string like |warehouse/*

Query 4
Query 5

1
SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL
Ln 1, Col 49

Run again
Explain
Cancel
Clear
Create

Reuse query results
up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 100 ms
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#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

icebreg_role
Info
Delete

Summary
Edit

Creation date
July 30, 2026, 09:56 (UTC+05:30)

ARN
arn:aws:iam::381492064215:role/icebreg_role

Link to switch roles in console
https://signin.aws.amazon.com/switchrole?roleName=icebreg_role&account=381492064215

Last activity
-

Maximum session duration
1 hour

Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions

Permissions policies (1)
Info
Simulate
Remove
Add permissions

You can attach up to 10 managed policies.

Filter by Type
All types

Policy name	Type	Attached entities
icebreg_ta_policy	Customer managed	1

4. Catalog Setup (AWS Glue - Single Shared Catalog)

Step 4.1: Create Glue Database (Namespace)

Glue Database Name	poc_db
Location URI	s3://ta-icebreg-poc/warehouse/poc_db

Screenshot Reference:

Query 4 : XQuery 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run againExplainCancelClearCreate

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)CopyDownload results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

AWS Glue > Databases > poc_db

Announcing new optimization features for Apache Iceberg tables
Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. [Learn more](#)

poc_dbLast updated (UTC)
August 1, 2026 at 08:36:22EditDelete

Database properties

Name poc_db	Description Iceberg cross-platform POV - Snowflake/Databricks/Athena	Location s3://ta-icebreg-poc/warehouse/poc_db	Created on (UTC) July 30, 2026 at 05:02:13
----------------	--	---	---

Tables (0)Last updated (UTC)
August 1, 2026 at 08:36:25DeleteAdd tables using crawlerAdd table

View and manage all available tables.

Filter tables

Name	Database	Location	Classific...	Depreca...	View data	Data quality
No available tables						

Step 4.2: Configure Glue Permissions

IAM Role Used	iceberg-poc-access-role
Lake Formation Settings (if applicable)	Lake Formation present but running in default IAM_ALLOWED_PRINCIPALS mode (no active admins configured) - access is governed entirely by IAM policy

Command Used: **aws lakeformation get-data-lake-settings** in **AWS** Cloud shell

Query 4

Query 5

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 100 ms

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Results (1)

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Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

Screenshot Reference :

```

~ $ aws lakeformation get-data-lake-settings
{
  "DataLakeSettings": {
    "DataLakeAdmins": [],
    "ReadOnlyAdmins": [],
    "CreateDatabaseDefaultPermissions": [
      {
        "Principal": {
          "DataLakePrincipalIdentifier": "IAM_ALLOWED_PRINCIPALS"
        },
        "Permissions": [
          "ALL"
        ]
      }
    ],
    "CreateTableDefaultPermissions": [
      {
        "Principal": {
          "DataLakePrincipalIdentifier": "IAM_ALLOWED_PRINCIPALS"
        },
        "Permissions": [
          "ALL"
        ]
      }
    ],
    "Parameters": {
      "CROSS_ACCOUNT_VERSION": "4",
      "SET_CONTEXT": "TRUE"
    },
    "TrustedResourceOwners": [],
    "AllowExternalDataFiltering": false,
    "ExternalDataFilteringAllowList": []
  }
}

```

5. Iceberg Table Creation

Step 5.1: Create Iceberg Table (registered in Glue)

Table Name	orders_iceberg
Schema Summary	order_id (int), customer_name (string), order_date (date), amount (double), status (string)
Created Via (Engine)	Athena
Partitioning	None

Command Used:

```

CREATE TABLE poc_db.orders_iceberg (
  order_id int,
  customer_name string,
  order_date date,

```

Query 4
Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again
Explain
Cancel
Clear
Create

Reuse query results up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 100 ms
Run time: 587 ms
Data scanned: 0.59 KB

Results (1)
Copy
Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

```

amount double,
status string
)
LOCATION 's3://your-org-iceberg-poc/warehouse/poc_db/orders_iceberg/'
TBLPROPERTIES (
  'table_type' = 'ICEBERG',
  'format' = 'parquet'
);

```

Screenshot Reference :

AWS Glue
Tables
orders_iceberg

Table overview
Data quality - new

Table details

Name
orders_iceberg

Database
poc_db

Description
-

Last updated
August 1, 2026 at 09:07:05

Table format
Apache Iceberg

Compaction status
Off

Snapshot retention status
Off

Orphan file deletion status
Off

Classification
-

Location
s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg

Connection
-

Deprecated
-

Column statistics
No statistics

Step 5.2: Load Sample Data

Record Count	5
Source of Sample Data	Manually inserted sample order records via Athena SQL

Query 4 : XQuery 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

Completed

Time in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

CopyDownload results CSV

Search rows

< 1 >

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Command / Config Used:

```
INSERT INTO poc_db.orders_iceberg VALUES
(1, 'Alice', DATE '2026-07-01', 120.50, 'pending'),
(2, 'Bob', DATE '2026-07-02', 89.99, 'shipped'),
(3, 'Charlie', DATE '2026-07-03', 45.00, 'pending'),
(4, 'Dana', DATE '2026-07-04', 210.75, 'delivered'),
(5, 'Evan', DATE '2026-07-05', 15.20, 'pending');
```

Screenshot Reference :

Data present in table

Query resultsQuery stats

Completed

Time in queue: 94 msRun time: 785 msData scanned: 0.36 KB

Results (5)

CopyDownload results CSV

Search rows

< 1 >

#	order_id	customer_name	order_date	amount	status
1	1	Alice	2026-07-01	120.5	pending
2	2	Bob	2026-07-02	89.99	shipped
3	3	Charlie	2026-07-03	45.0	pending
4	4	Dana	2026-07-04	210.75	delivered
5	5	Evan	2026-07-05	15.2	pending

Snapshot created

File created in S3 bucket

Query 4 : X

Query 5 : X

(+)

1

SELECT * FROM poc_db."orders_iceberg\$snapshots"

SQLLn 1, Col 49

Run again

▼

Explain

Cancel

Clear

Create

▼

Reuse query results

up to 60 minutes ago

Query resultsQuery stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Search rows

Copy

Download results CSV

<

1

>

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Amazon S3

>

Buckets

>

ta-icebreg-poc

>

warehouse/

>

poc_db/

>

orders_iceberg/

Amazon S3

<

▼ Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

▼ Files

File systems

▼ Access management and security

Access Points

Access Points for FSx

orders_iceberg/

Objects

Properties

Objects (2)

Objects are the fundamental entities stored in Amazon S3. For more information on object permissions, see [Object permissions](#).

Find objects by prefix

☐

Name

☐

data/

☐

metadata/

Query 4 X Query 5 X

1 SELECT * FROM poc_db.'orders_iceberg\$snapshots'

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

6. Platform Connections

6.1 Snowflake

Snowflake Trail Account	User Name	Password
Icebreg Trial Account	Tiger	Tiger123456789

Step 6.1.1: Create External Volume

External Volume Name	iceberg_poc_vol
STORAGE AWS IAM USER ARN	arn:aws:iam::008349342067:user/j5kw1000-s
STORAGE AWS EXTERNAL ID	iceberg_poc_ext_id
STORAGE AWS ROLE ARN	arn:aws:iam::381492064215:role/icebreg_role

Command Used:

```
CREATE OR REPLACE EXTERNAL VOLUME iceberg_poc_vol
  STORAGE_LOCATIONS = (
    (
      NAME = 'ta-icebreg-poc-s3'
      STORAGE_PROVIDER = 'S3'
      STORAGE_BASE_URL = 's3://ta-icebreg-poc/warehouse/poc_db/'
      STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::381492064215:role/icebreg_role'
      STORAGE_AWS_EXTERNAL_ID = 'iceberg_poc_ext_id'
    )
  )ALLOW_WRITES = TRUE;

DESC EXTERNAL VOLUME iceberg_poc_vol;

SELECT SYSTEM$VERIFY_EXTERNAL_VOLUME('iceberg_poc_vol');
```

Query 4 : X

Query 5 : X

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again

Explain

Cancel

Clear

Create

☐ Reuse query results
up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV

< 1 >

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Query 4 : X Query 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots"

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results

Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

```

1 CREATE OR REPLACE EXTERNAL VOLUME iceberg_poc_vol
2 STORAGE_LOCATIONS = (
3   (
4     NAME = 'ta-icebreg-poc-s3'
5     STORAGE_PROVIDER = 'S3'
6     STORAGE_BASE_URL = 's3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/'
7     STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::381492064215:role/ta-icebreg-poc-iceberg-s3-role'
8     STORAGE_AWS_EXTERNAL_ID = 'iceberg_poc_ext_id'
9   )
10 );
11
12 DESC EXTERNAL VOLUME iceberg_poc_vol;
13
14 SELECT SYSTEM$VERIFY_EXTERNAL_VOLUME('iceberg_poc_vol');
15

```

Results (just now)

Table

Chart

Pivot

1

SYSTEM\$VERIFY_EXTERNAL_VOLUME('ICEBERG_POC_VOL')

SYSTEM\$VERIFY_EXTERNAL_VOLUME('ICEBERG_POC_VOL')

1

{"success":true,"storageLocationSelectionResult":"PASSED"}

{"success":true,"storageLocationSelectionResult":"PASSED"}

s3", "servicePrincipalName": "ta-icebreg-poc-iceberg-s3-role",
 STORAGE_AWS_ROLE_ARN: "arn:aws:iam::381492064215:role/ta-icebreg-poc-iceberg-s3-role",
 STORAGE_AWS_EXTERNAL_ID: "iceberg_poc_ext_id",
 STORAGE_BASE_URL: "s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/",
 STORAGE_PROVIDER: "S3",
 NAME: "ta-icebreg-poc-s3",
 "writeResult": "PASSED",
 "azureGenerated": false

Query 4
Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again
Explain
Cancel
Clear
Create

Reuse query results
up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 100 ms
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Data scanned: 0.59 KB

Results (1)
Copy
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Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Notes / Issues Encountered:

Initial `SYSTEM$VERIFY_EXTERNAL_VOLUME` failed with `AccessDenied` on `s3:DeleteObject` — IAM policy on `icebreg_role` was missing delete permission on the `warehouse/*` prefix. Fixed by adding `s3:DeleteObject` to the policy; verification passed after update.

Step 6.1.2: Create Catalog Integration (CATALOG_SOURCE = GLUE)

Integration Name	glue_rest_catalog_int
Catalog Source	Icebreg REST
Glue Catalog ID	381492064215
Glue Region	eu-north-1
Glue AWS Role ARN	arn:aws:iam::381492064215:role/icebreg_role
Table Format	Iceberg

Command Used:

```
CREATE OR REPLACE CATALOG INTEGRATION glue_rest_catalog_int
  CATALOG_SOURCE = ICEBERG_REST
  TABLE_FORMAT = ICEBERG
  CATALOG_NAMESPACE = 'poc_db'
  REST_CONFIG = (
    CATALOG_URI = 'https://glue.eu-north-1.amazonaws.com/iceberg'
    CATALOG_API_TYPE = AWS_GLUE
    CATALOG_NAME = '381492064215'
  )
  REST_AUTHENTICATION = (
    TYPE = SIGV4
    SIGV4_IAM_ROLE = 'arn:aws:iam::381492064215:role/icebreg_role'
    SIGV4_SIGNING_REGION = 'eu-north-1'
  )
```

Query 4 : XQuery 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

```
ENABLED = TRUE;  
  
DESC CATALOG INTEGRATION glue_catalog_int;  
  
SELECT SYSTEM$VERIFY_CATALOG_INTEGRATION('glue_rest_catalog_int');
```

Screenshot Reference :

```
CREATE OR REPLACE CATALOG INTEGRATION glue_rest_catalog_int  
CATALOG_SOURCE = ICEBERG_REST  
TABLE_FORMAT = ICEBERG  
CATALOG_NAMESPACE = 'poc_db'  
REST_CONFIG = (  
  CATALOG_URI = 'https://glue.eu-north-1.amazonaws.com/iceberg'  
  CATALOG_API_TYPE = AWS_GLUE  
  CATALOG_NAME = '381492064215'  
)  
REST_AUTHENTICATION = (  
  TYPE = SIGV4  
  SIGV4_IAM_ROLE = 'arn:aws:iam::381492064215:role/icebreg_role'  
  SIGV4_SIGNING_REGION = 'eu-north-1'  
)  
ENABLED = TRUE;  
  
DESC CATALOG INTEGRATION glue_rest_catalog_int;
```

ts (just now)

TableChartPivot

10 rows1.2s

property	property_type	property_value	property_default
APL_AWS_EXTERNAL_ID	String		
APL_AWS_IAM_USER_ARN	EnumMap	30	30
+8 more	+2 more	+8 more	+1 more
ENABLED	Boolean	true	true
APL_AWS_IAM_USER_ARN	String	arn:aws:iam::008349342067:user/j5kw100	
APL_AWS_EXTERNAL_ID	String	TX33704_SFCRole=5_/F22x6iuM/FEZ8MTa	
CATALOG_SOURCE	String	ICEBERG_REST	
CATALOG_NAMESPACE	String	poc_db	
TABLE_FORMAT	String	ICEBERG	
REFRESH_INTERVAL_SECONDS	Integer	30	30

Notes / Issues Encountered:

Query 4 : X Query 5 : X

1 SELECT * FROM poc_db.'orders_iceberg\$snapshots'

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

1. Catalog integration uses a **separate** IAM principal from the external volume — required a second **Statement** block in **icebreg_role**'s trust policy (initial **AssumeRole: 403** error until this was added).
2. **SYSTEM\$VERIFY_CATALOG_INTEGRATION** is **not supported for GLUE catalog type** (**errorCode 004186**) — this is expected behavior, not a config error. Verification was instead done by directly querying the registered table (see 6.1.3).

Step 6.1.3: Query Table from Snowflake

Query Used	<pre>CREATE ICEBERG TABLE POC_DB.DEV.orders_iceberg EXTERNAL_VOLUME = 'iceberg_poc_vol' CATALOG = 'glue_rest_catalog_int' CATALOG_TABLE_NAME = 'orders_iceberg' AUTO_REFRESH = TRUE; SELECT * FROM orders_iceberg;</pre>
Result Summary	All 5 rows originally inserted via Athena are visible and match exactly — confirms Snowflake can read a Glue-cataloged Iceberg table with no data duplication
Read or Read-Write?	Confirmed read; Write capability to be confirmed in Section 7 (INSERT sync test)

Screenshot Reference :

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run againExplainCancelClearCreate

Reuse query results up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)CopyDownload results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

```
CREATE ICEBERG TABLE POC_DB.DEV.orders_iceberg
EXTERNAL_VOLUME = 'iceberg_poc_vol'
CATALOG = 'glue_rest_catalog_int'
CATALOG_TABLE_NAME = 'orders_iceberg';

-----

-- STEP 4: Test READ

-----

SELECT * FROM POC_DB.DEV.orders_iceberg;
```

Results (just now)

TableChartPivot

#	ORDER_ID	CUSTOMER_NAME
	8	Henry
	1	Alice
	2	Bob
	3	Charlie

Query 4

Query 5

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV

Search rows

#

committed_at

snapshot_id

parent_id

operation

manifest_list

1

2026-08-01 09:07:04.968 UTC

952695883030641274

append

s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

6.2 Databricks

Step 6.2.1: Configure Glue as Foreign Catalog

Approach Used	Hive Metastore Federation → AWS Glue (Foreign Catalog)
IAM Role	icebreg_role
Service Credential External ID	e34417ac-99f0-4aaa-8d76-965a32255318 (Databricks)
AWS Region	eu-north-1
Foreign Catalog Name	glue poc catalog

Command /Used:

```
CREATE CONNECTION db_glue_catalog TYPE glue
OPTIONS (aws_region 'eu-north-1', aws_account_id '381492064215', credential
'glue_poc_credential');

CREATE FOREIGN CATALOG db_glue_catalog_catalog USING CONNECTION db_glue_catalog
OPTIONS (authorized_paths 's3://ta-iceberg-poc/');
```

Screenshot Reference :

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

CopyDownload results CSV

Search rows

<1>

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Catalog Explorer >

glue_poc_catalog

OverviewDetailsPermissionsPoliciesWorkspaces

Description

AI generate

Add

Filter schemas 1 schema	
Name	Owner
poc_db	aslam26

Query 4 : X Query 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1) Copy Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

Catalog Explorer > glue_poc_catalog >

 **poc_db**  

Overview Details Permissions Policies

Description 

Iceberg cross-platform POV - Snowflake/Da

 Filter tables

Query 4 : X Query 5 : X

1 SELECT * FROM poc_db.`orders\_iceberg\$snapshots`

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

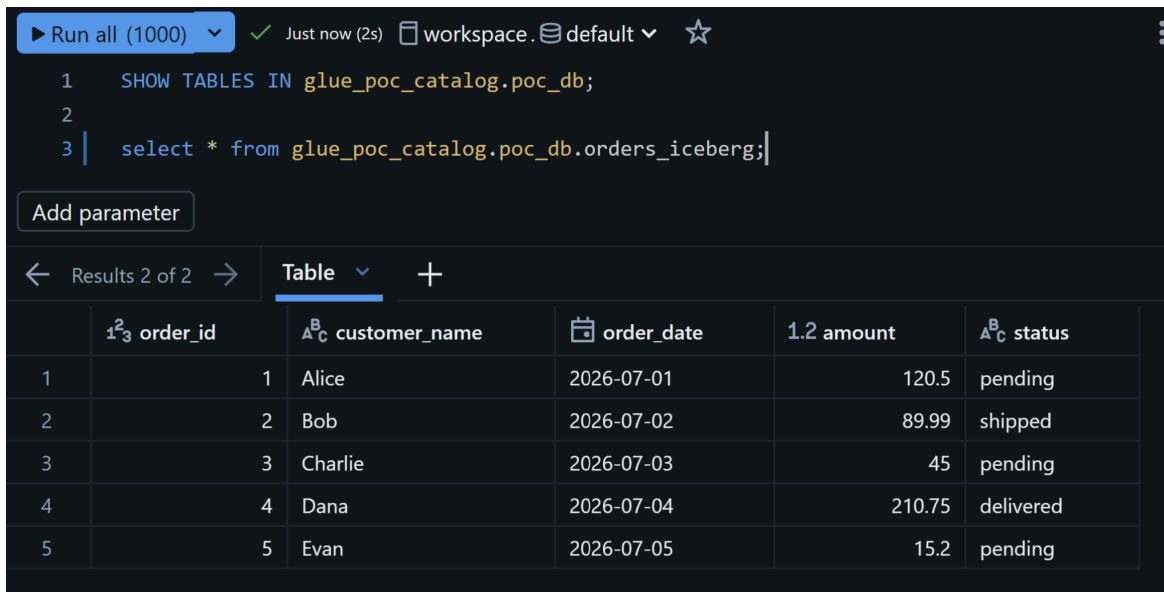
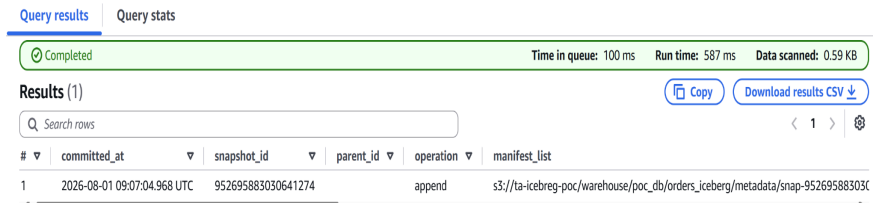
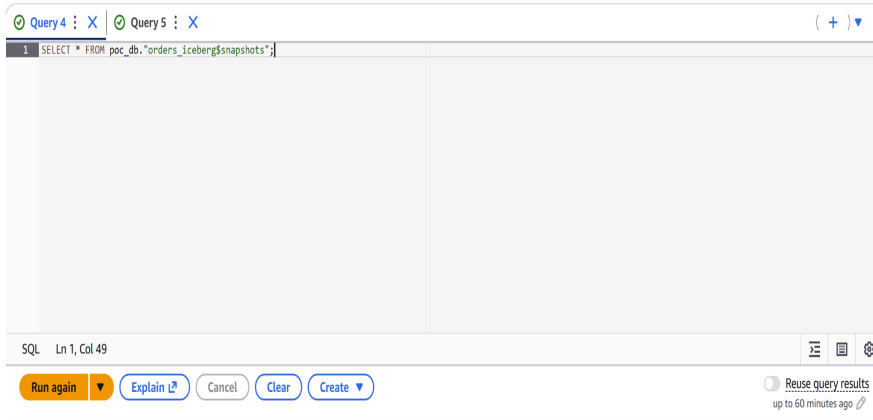
Notes / Issues Encountered:

1. Workspace is serverless-only (trial) — no cluster/compute option available, so pivoted from the originally planned Spark-config approach to Unity Catalog's native Glue Foreign Catalog feature.
2. Initial catalog creation failed with **UC_IAM_ROLE_NON_SELF_ASSUMING** — the storage credential's IAM role needs a trust statement allowing it to assume itself (an AWS-wide requirement since June 2023). Fixed by adding a self-referencing trust statement.

Step 6.2.2: Query Table from Databricks

Query Used	REFRESH FOREIGN CATALOG db_glue_catalog_catalog; SHOW SCHEMAS IN db_glue_catalog_catalog; SHOW TABLES IN db_glue_catalog_catalog.poc_db; SELECT * FROM db_glue_catalog_catalog.poc_db.orders_iceberg;
Result Summary	all 5 rows visible / matches Athena & Snowflake exactly
Read or Read-Write?	Confirmed read; write capability to be tested in Section 7

Screenshot Reference :



6.3 Amazon Athena

Step 6.3.1: Confirm Glue Database Visible in Athena

Workgroup Used	Primary
Query Engine Version	Athena engine version 3
Query Used	SELECT * FROM poc_db.orders_iceberg;
Result Summary	All 5 rows visible — same as originally loaded; confirms baseline before cross-platform sync tests begin
Read or Read-Write?	Confirmed read-write

Screenshot Reference :

Query 4

Query 5

1

SELECT * FROM poc_db."orders_iceberg\$snapshots"

SQL

Ln 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV

Search rows

#

committed_at

snapshot_id

parent_id

operation

manifest_list

1

2026-08-01 09:07:04.968 UTC

952695883030641274

append

s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Query results

Query stats

Completed

Time in queue: 94 ms

Run time: 785 ms

Data scanned: 0.36 KB

Results (5)

Copy

Download results CSV

Search rows

#

order_id

customer_name

order_date

amount

status

1

1

Alice

2026-07-01

120.5

pending

2

2

Bob

2026-07-02

89.99

shipped

3

3

Charlie

2026-07-03

45.0

pending

4

4

Dana

2026-07-04

210.75

delivered

5

5

Evan

2026-07-05

15.2

pending

Notes / Issues Encountered:

Initial setup required configuring a query result output location

(<s3://ta-iceberg-poc-athena-results/>) — Athena requires this separately from the Iceberg warehouse path itself, since it stores query metadata/logs there rather than table data.

7. Cross-Platform Sync Proof Tests

For each test, run the operation on the "Source" platform, then immediately query from the other two platforms and record what each one sees. Note any delay before the change becomes visible.

7.1 INSERT Sync Test

Step 7.1.1: Insert from Snowflake -> Verify in Databricks & Athena

Rows Inserted	1
Visible in Databricks? (Y/N + delay)	
Visible in Athena? (Y/N + delay)	Yes

Query 4 : XQuery 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

Completed

Time in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.m

Command / Config Used:

```
INSERT INTO POC_DB.DEV.orders_iceberg VALUES
(6, 'Fiona', DATE '2026-08-01', 75.00, 'pending', null);
```

Output / Confirmation:

INSERT INTO POC_DB.DEV.orders_iceberg VALUES
(6, 'Fiona', DATE '2026-08-01', 75.00, 'pending', null);

SELECT * FROM POC_DB.DEV.orders_iceberg;

just now

ChartPivot

1 row4.7s

number of rows inserted

1

Screenshot Reference:

SELECT * FROM POC_DB.DEV.orders_iceberg;

just now

ChartPivot

7 rows

ORDER_ID	CUSTOMER_NAME	ORDER_DATE	AMOUNT	STATUS
8	Henry	2026-08-01	99.99	pending
6	Fiona	2026-08-01	75	pending
1	Alice	2026-07-01	120.5	pending
2	Bob	2026-07-02	89.99	shipped
3	Charlie	2026-07-03	45	pending
4	Dana	2026-07-04	210.75	delivered
5	Evan	2026-07-05	15.2	pending

Query 4
Query 5

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again
Explain
Cancel
Clear
Create

Reuse query results up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 100 ms
Run time: 587 ms
Data scanned: 0.59 KB

Results (1)
Copy
Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Query 9
Query 4
Query 5
Query 6
Query 7
Query 8

SELECT * FROM poc_db.orders_iceberg WHERE order_id = 6;

SQL Ln 1, Col 56

Run again
Explain
Cancel
Clear
Create

Reuse query results up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 62 ms
Run time: 1.063 sec
Data scanned: 0.23 KB

Results (1)
Copy
Download results CSV

Search rows

#	order_id	customer_name	order_date	amount	status	region
1	6	Fiona	2026-08-01	75.0	pending	

Step 7.1.2: Insert from Databricks -> Verify in Snowflake & Athena

Rows Inserted	
Visible in Snowflake? (Y/N + delay)	
Visible in Athena? (Y/N + delay)	

Command / Config Used:

Output / Confirmation:

Screenshot Reference (filename):

e.g., 01-s3-bucket-created.png

The screenshot shows a SQL query execution interface. The query is: `SELECT * FROM poc_db."orders_iceberg$snapshots";`. The query is labeled "Query 4" and "Query 5". The status is "Completed". The query statistics are: Time in queue: 100 ms, Run time: 587 ms, Data scanned: 0.59 KB. The results are shown in a table with 1 row.

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Notes / Issues Encountered:

Step 7.1.3: Insert from Athena -> Verify in Snowflake & Databricks

Rows Inserted	1
Visible in Snowflake? (Y/N + delay)	Yes
Visible in Databricks? (Y/N + delay)	

Command / Config Used:

```
INSERT INTO poc_db.orders_iceberg VALUES
(8, 'Henry', DATE '2026-08-01', 99.99, 'pending', 'EU');
```

Output / Confirmation:

The screenshot shows a SQL query execution interface. The query is: `INSERT INTO poc_db.orders_iceberg VALUES (8, 'Henry', DATE '2026-08-01', 99.99, 'pending', 'EU');`. The query is labeled "Query 9", "Query 6", "Query 7", "Query 8", and "Query 4". The status is "Completed". The query statistics are: Time in queue: 44 ms, Run time: 1.202 sec, Data scanned: -. The results are shown in a table with 1 row.

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Screenshot Reference :

Query 4 : X
Query 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Search rows

committed_at snapshot_id parent_id operation manifest_list
1 2026-08-01 09:07:04.968 UTC 952695883030641274 append s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

Query 9 : X Query 6 : X Query 7 : X Query 8 : X Query 4 : X

1 SELECT * FROM poc_db.orders_iceberg WHERE order_id = 8;

SQL Ln 1, Col 55

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 103 ms Run time: 1.129 sec Data scanned: 0.25 KB

Results (1)

Search rows

order_id customer_name order_date amount status region
1 8 Henry 2026-08-01 99.99 pending EU

12 SELECT * FROM POC_DB.DEV.orders_iceberg;

Results (2 minutes ago)

Table Chart Pivot

7 rows 57ms

ORDER_ID CUSTOMER_NAME ORDER_DATE AMOUNT STATUS REGION
1 8 Henry 2026-08-01 99.99 pending EU
2 6 Fiona 2026-08-01 75 pending null
3 1 Alice 2026-07-01 120.5 pending null
4 2 Bob 2026-07-02 89.99 shipped null
5 3 Charlie 2026-07-03 45 pending null
6 4 Dana 2026-07-04 210.75 delivered null
7 5 Evan 2026-07-05 15.2 pending null

7.2 UPDATE Sync Test

Step 7.2.1: Update from Snowflake -> Verify in Databricks & Athena

Rows Updated	1
Visible in Databricks? (Y/N + delay)	
Visible in Athena? (Y/N + delay)	

Command / Config Used:

```
UPDATE POC_DB.DEV.orders_iceberg
SET status = 'shipped', amount = 80.00
```

Query 4 : XQuery 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

WHERE order_id = 6;

Output / Confirmation:

```
PREREQ: poc_db_dev orders_iceberg
SET status = 'shipped', amount = 80.00
WHERE order_id = 6;

SELECT * FROM poc_db_dev orders_iceberg WHERE order_id = 6;

-- STEP 7: Test DELETE
DEL FROM poc_db_dev orders_iceberg
WHERE order_id = 6;
```

#	number of rows updated	number of multi-part rows updated
1	1	0

Screenshot Reference:

404142SELECT * FROM poc_db_dev orders_iceberg WHERE order_id = 6;

Results (just row)

#	order_id	customer_name	order_date	amount	status	region
1	6	Fiona	2026-08-01	80.0	shipped	us

Query 9 : XQuery 6 : XQuery 7 : XQuery 8 : XQuery 4 : X

1 SELECT * FROM poc_db.orders_iceberg WHERE order_id = 6;

SQLLn 1, Col 55

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 66 msRun time: 1.091 secData scanned: 0.23 KB

Results (1)

Search rows

CopyDownload results CSV

#	order_id	customer_name	order_date	amount	status	region
1	6	Fiona	2026-08-01	80.0	shipped	us

Step 7.2.2: Update from Databricks -> Verify in Snowflake & Athena

Rows Updated	
--------------	--

Query 4 : X

Query 5 : X

(+) ▼

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

▼

Explain L2

Cancel

Clear

Create ▼

Reuse query results

up to 60 minutes ago

Query resultsQuery stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Search rows

Copy

Download results CSV

< 1 >

⚙

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

Visible in Snowflake? (Y/N + delay)	
Visible in Athena? (Y/N + delay)	

Command / Config Used:

Output / Confirmation:

Screenshot Reference (filename):

e.g., 01-s3-bucket-created.png

Notes / Issues Encountered:

Step 7.2.3: Update from Athena -> Verify in Snowflake & Databricks

Rows Updated	1
Visible in Snowflake? (Y/N + delay)	Yes (delay ~30s)
Visible in Databricks? (Y/N + delay)	

Command / Config Used:

```
UPDATE poc_db.orders_iceberg
```

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

SET status = 'shipped'
WHERE order_id = 8;

Output / Confirmation:

Query 4 : X Query 5 : X (+) ▼

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again Explain L Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Query 9 : X Query 6 : X Query 7 : X

```
1 UPDATE poc_db.orders_iceberg
2 SET status = 'shipped'
3 WHERE order_id = 8;
4
5
```

SQL Ln 5, Col 1

Run again ▼

Explain ↗

Cancel

Clear

Query results

Query stats

Completed

Query successful.

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

Screenshot Reference :

Query 4 : X | Query 5 : X | (+) ▼

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again ▼ | Explain ↗ | Cancel | Clear | Create ▼

Reuse query results up to 60 minutes ago

Query results | Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1) Copy Download results CSV ↗

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

Query 9 : X | Query 6 : X | Query 7 : X

1 SELECT * FROM poc_db.orders_iceberg WHERE order_id = 1;

SQL Ln 1, Col 54

Run again ▼ | Explain ↗ | Cancel | Clear

Query results | Query stats

Completed

Results (1)

Search rows

#	order_id	customer_name
---	----------	---------------

Query 4 : XQuery 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

SELECT * FROM POC_DB.DEV.orders_iceberg;

Results (just now)

Table

Chart

Pivot

#	ORDER_ID	CUSTOMER_NAME
	6	Fiona
	8	Henry
	1	Alice
	2	Bob
	3	Charlie
	4	Dana
	5	Evan

Notes / Issues Encountered:

Query 4
Query 5

1
SELECT * FROM poc_db."orders_iceberg\$snapshots"

SQL
Ln 1, Col 49

Run again
Explain
Cancel
Clear
Create

Reuse query results
up to 60 minutes ago

Query results
Query stats

Completed
Time in queue: 100 ms
Run time: 587 ms
Data scanned: 0.59 KB

Results (1)
Copy
Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

Table uses the Glue Iceberg REST catalog integration, it should auto-refresh within ~30 seconds.

7.3 DELETE Sync Test

Step 7.3.1: Delete from Snowflake -> Verify in Databricks & Athena

Rows Deleted	1
Visible in Databricks? (Y/N + delay)	
Visible in Athena? (Y/N + delay)	Yes

Command / Config Used:

```
DELETE FROM POC_DB.DEV.orders_iceberg
WHERE order_id = 6;
```

Output / Confirmation:

Query 4 : X

Query 5 : X

(+) ▼

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again ▼

Explain ↗

Cancel

Clear

Create ▼

Reuse query results

up to 60 minutes ago ↗

Query resultsQuery stats

Completed

Time in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

CopyDownload results CSV ↗

Search rows

< 1 >

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1-1.parquet

DELETE FROM POC_DB.DEV.orders_iceberg

WHERE order_id = 6;

SELECT * FROM POC_DB.DEV.orders_iceberg;

(just now)

ChartPivot

number of rows deleted

Screenshot Reference :

Query 4 : X

Query 5 : X

(+) ▼

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

▼

Explain ↗

Cancel

Clear

Create ▼

Reuse query results

up to 60 minutes ago ↗

Query resultsQuery stats

Completed

Time in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV ↗

< 1 > ⚙

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

9

0

1

SELECT * FROM POC_DB.DEV.orders_iceberg WHERE order_id = 6;

Results (just now)

Table

Chart

Pivot

	ORDER_ID	CUSTOMER_NAME
--	----------	---------------

Query 4 : X

Query 5 : X

(+)

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again

▼

Explain ↗

Cancel

Clear

Create ▼

Reuse query results

up to 60 minutes ago

Query results | Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV ↗

Search rows

<

1

>

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

Query 9 : X

Query 6 : X

Query 7 : X

1

SELECT * FROM poc_db.orders_iceberg WHERE order_id = 6;

SQL Ln 1, Col 55

Run again

▼

Explain ↗

Cancel

Clear

Query results | Query stats

Completed

Results (0)

Search rows

#	order_id	customer_name
---	----------	---------------

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

Step 7.3.2: Delete from Databricks -> Verify in Snowflake & Athena

Rows Deleted	
Visible in Snowflake? (Y/N + delay)	
Visible in Athena? (Y/N + delay)	

Command / Config Used:

Output / Confirmation:

Screenshot Reference (filename):

e.g., 01-s3-bucket-created.png

Notes / Issues Encountered:

Step 7.3.3: Delete from Athena -> Verify in Snowflake & Databricks

Rows Deleted	1
Visible in Snowflake? (Y/N + delay)	Yes

Query 4 : X

Query 5 : X

(+)

▼

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL

Ln 1, Col 49

Run again

▼

Explain

Cancel

Clear

Create

▼

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV

Search rows

< 1 >

⚙

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Visible in Databricks?
(Y/N + delay)

Command / Config Used:

```
DELETE FROM poc_db.orders_iceberg WHERE order_id = 8;
```

Output / Confirmation:

Query 9 : X

Query 6 : X

Query 7 : X

Query 10 : X

Query 8 : X

Query 4 : X

(+)

▼

1

delete FROM poc_db.orders_iceberg WHERE order_id = 8;

SQL

Ln 1, Col 54

Run again

▼

Explain

Cancel

Clear

Create

▼

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 46 ms

Run time: 1.235 sec

Data scanned: 0.04 KB

Query successful.

Screenshot Reference :

Query 9 : X

Query 6 : X

Query 7 : X

Query 10 : X

Query 8 : X

Query 4 : X

(+)

▼

1

SELECT * FROM poc_db.orders_iceberg WHERE order_id = 8;

SQL

Ln 1, Col 55

Run again

▼

Explain

Cancel

Clear

Create

▼

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 127 ms

Run time: 2.596 sec

Data scanned: -

Results (0)

Copy

Download results CSV

Search rows

< 1 >

⚙

#	order_id	customer_name	order_date	amount	status	region
No results						
Run a query to view results						

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run againExplainCancelClearCreate

Reuse query results up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)CopyDownload results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.m

SELECT * FROM POC_DB.DEV.orders_iceberg WHERE order_id = 8;

Ctrl+I to generate

Results (1 minute ago)

TableChartPivot

0 rows146ms

ORDER_ID	CUSTOMER_NAME	ORDER_DATE	AMOUNT	STATUS	REGION
----------	---------------	------------	--------	--------	--------

Query produced no results

7.4 Additional Checks

Step 7.4.1.1: Schema Evolution Test (Snowflake)

Change Made	Snowflake
Visible in Other Platforms? (Y/N)	Yes Visible in

Command / Config Used:

```
ALTER ICEBERG TABLE POC_DB.DEV.orders_iceberg DROP COLUMN region;
```

Output / Confirmation:

Query 4 : X

Query 5 : X

(+) ▼

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again ▼

Explain L2

Cancel

Clear

Create ▼

Reuse query results

up to 60 minutes ago ↗

Query resultsQuery stats

Completed

Time in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

CopyDownload results CSV ↗

Search rows

< 1 >

⚙

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

✦

ALTER ICEBERG TABLE POC_DB.DEV.orders_iceberg\$snapshots

its (just now)

Table

Chart

Pivot

A status

Statement executed successfully.

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Screenshot Reference (filename):

Query 4 : X | Query 5 : X | (+) ▼

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again ▼ Explain ↗ Cancel Clear Create ▼

Reuse query results up to 60 minutes ago

Query results | Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1.parquet

❌ Query 9 : X | ✅ Query 6 : X | ✅ Query 7 : X |

1 SELECT region FROM poc_db.orders_iceberg;

SQL Ln 1, Col 14

Run again ▼ Explain ↗ Cancel Clear

Query results | Query stats

❌ Failed

❌ COLUMN_NOT_FOUND: line 1:8: Column 'region' cannot be resolved. This query ran against the "poc_db" database, unless qualified with a schema name. See [support](#) with Query Id: bd54c396-e82e-46c5-a805-9a8e38

Query 4 : X

Query 5 : X

(+) ▼

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

▼

Explain ↗

Cancel

Clear

Create ▼

Reuse query results

up to 60 minutes ago ↗

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV ↗

Search rows

#

▼

committed_at

▼

snapshot_id

▼

parent_id

▼

operation

▼

manifest_list

1

2026-08-01 09:07:04.968 UTC

952695883030641274

append

s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

Notes / Issues Encountered:

Step 7.4.1.2: Schema Evolution Test (Athena)

Change Made	Athena
Visible in Other Platforms? (Y/N)	Yes Visible in Snowflake

Command / Config Used:

```
ALTER TABLE poc_db.orders_iceberg ADD COLUMNS (region string);
```

Output / Confirmation:

Query 4 : X | Query 5 : X

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results | Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1) Copy Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Query 9 : X | Query 6 : X | Query 7 : X | Query 8 : X

1 ALTER TABLE poc_db.orders_iceberg ADD COLUMNS (region string, ...)

2

SQL Ln 1, Col 63

Run again Explain Cancel Clear

Query results | Query stats

Completed

Query successful.

Query 4 : X

Query 5 : X

(+) ▼

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again ▼

Explain ↗

Cancel

Clear

Create ▼

Reuse query results

up to 60 minutes ago ↗

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV ↗

Search rows

# ▼	committed_at ▼	snapshot_id ▼	parent_id ▼	operation ▼	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

Screenshot Reference (filename):

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

CopyDownload results CSV

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-icebreg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

SELECT region FROM POC_DB.DEV.orders_iceberg;

ALTER ICEBERG TABLE POC_DB.DEV.orders_iceberg DROP COLUMN region;

ts (just now)

Table

Chart

Pivot

REGION
null
null
null
null
null

Notes / Issues Encountered:

Query 4Query 5

1 SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

CompletedTime in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)

Search rows

#

committed_at

snapshot_id

parent_id

operation

manifest_list

12026-08-01 09:07:04.968 UTC952695883030641274appends3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

Schema changes executed in Athena are **not automatically reflected** in Snowflake in real time; they require a manual refresh of the Snowflake Iceberg table.

Step 7.4.1.3: Schema Evolution Test (Databricks)

Change Made	Databricks
Visible in Other Platforms? (Y/N)	

Command / Config Used:

```
ALTER TABLE poc_db.orders_iceberg ADD COLUMNS (region string);
```

Output / Confirmation:

Screenshot Reference ():

Notes / Issues Encountered:

Step 7.4.2: Time Travel / Snapshot Query (per platform)

Snapshot ID / Timestamp Used	
Result per Platform	

Command / Config Used:

Query 4Query 5

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274-1

Output / Confirmation:

Screenshot Reference (filename):

e.g., 01-s3-bucket-created.png

Notes / Issues Encountered:

Step 7.4.3: Concurrent Write Test

Platforms Involved	
Conflict Behavior Observed	

Command / Config Used:

Output / Confirmation:

Query 4Query 5

1

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again

Explain

Cancel

Clear

Create

Reuse query results

up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 100 ms

Run time: 587 ms

Data scanned: 0.59 KB

Results (1)

Copy

Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Screenshot Reference (filename):
e.g., 01-s3-bucket-created.png

Notes / Issues Encountered:

8. Findings & Observations

8.1 Platform Capability Matrix

Capability	Snowflake (Iceberg REST)	Databricks (Foreign Catalog)	Athena (Native Glue)
Read	Yes	Yes (breaks after V2 deletes)	Yes
INSERT	Yes	Not tested (blocked)	Yes
UPDATE	Yes	Not tested (blocked)	Yes
DELETE	Yes	Not tested (blocked)	Yes
MERGE	Yes	Not tested (blocked)	Yes
Schema Evolution (ADD/DROP column)	Yes (propagates to Glue)	Read-only (cannot alter)	Yes

Query 4
Query 5

SELECT * FROM poc_db."orders_iceberg\$snapshots";

SQLLn 1, Col 49

Run again
Explain
Cancel
Clear
Create

Reuse query results
up to 60 minutes ago

Query resultsQuery stats

Completed
Time in queue: 100 msRun time: 587 msData scanned: 0.59 KB

Results (1)
CopyDownload results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Auto-Refresh (sees other engines' writes)	Yes (~30s polling)	Yes (instant, reads live snapshot)	Yes (instant, reads live snapshot)
Time Travel	Limited (snapshot-based)	Limited (snapshot-based)	Yes (FOR SYSTEM_TIME AS OF)

8.2 What Worked Seamlessly

Single catalog, zero duplication: A single Iceberg table, created once via Athena and cataloged in AWS Glue, was successfully queried from all three platforms with no data copying or separate load step.

Snowflake full read-write via Iceberg REST: Using CATALOG_SOURCE = ICEBERG_REST (Glue's Iceberg REST endpoint), Snowflake achieved full DML capability — INSERT, UPDATE, DELETE, and schema evolution (DROP COLUMN) all worked and propagated back to Glue. This is a critical distinction from the legacy CATALOG_SOURCE = GLUE integration, which is read-only.

Athena native integration: Zero special configuration beyond the query-result output location. All DML and DDL operations worked with standard SQL.

Cross-platform sync for appends: INSERTs from any platform were visible on the others — Athena and Databricks instantly (they read live S3 metadata); Snowflake within ~30 seconds via auto-refresh polling.

Schema evolution propagation: Column additions (Athena) and column drops (Snowflake) were visible across platforms after metadata refresh, with no data rewrites required — Iceberg's column-ID-based schema tracking works as documented.

8.3 Friction Points & Workarounds

Issue	Platform	Impact
-------	----------	--------

Query 4 X Query 5 X

1 SELECT * FROM poc_db.'orders_iceberg\$snapshots'

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

IAM trust policy complexity	Snowflake	Multiple distinct external IDs required (one per integration object) – see trial-and-error
Lake Formation credential vending	Snowflake (REST)	Unable to assume role error blocked all access
Legacy vs. REST integration confusion	Snowflake	CATALOG_SOURCE = GLUE is read-only; documentation doesn't prominently mention this from REST
V2 delete-file incompatibility	Databricks	Any Iceberg merge-on-read DELETE file (produced by UPDATE/DELETE in Snowflake) renders the table completely unreadable from Databricks via assertion failure (DeleteFileWrapper)
Serverless-only trial workspace	Databricks	No classic cluster available – could not test Spark-config-based Iceberg
Self-assuming role requirement	Databricks	UC_IAM_ROLE_NON_SELF_ASSUMING error on catalog creation
Schema refresh not automatic	Snowflake	DDL changes from Athena not visible until refresh

8.4 Key Insight: The Databricks V2 Delete-File Problem

This is the most significant finding of this POV. Databricks' Foreign Catalog / UniForm Ingress cannot read Iceberg V2 positional delete files (the default write mode for UPDATE/DELETE in Athena and Snowflake). Once a single UPDATE or DELETE is performed from any other engine, Databricks loses read access entirely – not gracefully (no stale data), but with a hard failure.

Implications for production:

Query 4 : X Query 5 : X

```
1 SELECT * FROM poc_db.'orders_iceberg$snapshots'
```

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1)

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

- If Databricks is in the architecture, UPDATE/DELETE operations must either (a) be performed exclusively from Databricks itself, or (b) use copy-on-write mode (slower writes but produces only data files), or (c) periodic compaction must run from a non-Databricks engine to rewrite delete files into base data files.
- For read-only Databricks access to a table that receives UPDATES from other engines, this is a blocker without compaction automation.

8.5 Recommendations

1. For read-write multi-engine access (Snowflake + Athena): This pattern works well today. Use `CATALOG_SOURCE = ICEBERG_REST` in Snowflake with `AUTO_REFRESH = TRUE`. Both engines can INSERT, UPDATE, DELETE, and changes propagate within seconds.
2. For including Databricks in the architecture: Limit Databricks to append-only tables, or run Databricks as the sole writer (with others as readers). Alternatively, implement automated Iceberg compaction (via Athena `OPTIMIZE` or Glue Auto-Compaction) on a schedule to convert delete files before Databricks reads.
3. IAM setup: Budget extra time for IAM trust policy configuration. Document all external IDs upfront via `DESC EXTERNAL VOLUME` and `DESC CATALOG INTEGRATION` before touching AWS.
4. Lake Formation: If you don't need fine-grained column/row-level security, consider disabling Lake Formation governance on the database (use IAM-only mode) to avoid the credential-vending complexity entirely.

8.6 Final Verdict

Question	Answer
Can all three platforms read the same table?	Yes — but Databricks breaks after any V2 delete operation from another engine
Can all three platforms write?	Snowflake + Athena: Yes. Databricks: Read-only (Foreign Catalog limitation)
How fast is cross-platform sync?	Athena/Databricks: instant. Snowflake: ~30 seconds with auto-refresh
Where does it break down?	Databricks V2 delete-file incompatibility; Lake Formation credential vending complexity; IAM multi-principal trust setup

Query 4 X Query 5 X

1 SELECT * FROM poc_db.'orders_iceberg\$snapshots'

SQL Ln 1, Col 49

Run again Explain Cancel Clear Create

Reuse query results up to 60 minutes ago

Query results Query stats

Completed Time in queue: 100 ms Run time: 587 ms Data scanned: 0.59 KB

Results (1) Copy Download results CSV

Search rows

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Is this viable for production?

Yes, for Snowflake + Athena as a dual-engine pattern. Adding Databricks requires architectural constraints (append-only or compaction automation)

8. Findings & Observations

What Worked Seamlessly

- A single Iceberg table, created once via Athena and cataloged in AWS Glue, was successfully queried from all three platforms (Snowflake, Databricks, Athena) with zero data duplication — the core "write once, catalog once, read from anywhere" premise of this POV held true for reads.
- Athena required no special configuration beyond the standard query-result output location — it is Glue-native, so table creation, inserts, updates, and deletes all worked directly with standard SQL.
- Databricks' automatic sync for row-level INSERTs worked cleanly — new rows were visible immediately with zero manual refresh, reading the live Iceberg snapshot from S3 on every query.
- Cross-platform read access, once each connection was correctly configured, was reliable and fast for simple append-only data (inserts with no updates/deletes).

Friction Points / Workarounds Needed

- IAM/trust policy complexity: both Snowflake and Databricks required multiple, distinct IAM trust relationships on top of standard S3/Glue permissions — Snowflake needed separate principals for its External Volume and Catalog Integration; Databricks needed separate credentials for cross-account access (control plane) and storage/service access (data plane), each requiring self-assuming trust policies (an AWS-wide requirement since June 2023). This added significant setup time and several rounds of trial-and-error before each connection succeeded.
- Databricks trial workspace was serverless-only — no classic cluster/compute option was available, which ruled out the originally planned approach of setting custom Spark

The screenshot displays the Databricks SQL interface. At the top, there are tabs for 'Query 4' and 'Query 5'. The active query is 'Query 4', which contains the SQL statement: `SELECT * FROM poc_db."orders_iceberg$snapshots"`. Below the query editor, there are buttons for 'Run again', 'Explain', 'Cancel', 'Clear', and 'Create'. A status bar indicates 'SQL Ln 1, Col 49'. Below the query editor, there are tabs for 'Query results' and 'Query stats'. The 'Query results' tab is active, showing a green bar indicating 'Completed'. Below this, there are buttons for 'Copy' and 'Download results CSV'. The results table has columns: '#', 'committed_at', 'snapshot_id', 'parent_id', 'operation', and 'manifest_list'. The first row of data is:

#	committed_at	snapshot_id	parent_id	operation	manifest_list
1	2026-08-01 09:07:04.968 UTC	952695883030641274		append	s3://ta-iceberg-poc/warehouse/poc_db/orders_iceberg/metadata/snap-952695883030641274

Iceberg-catalog config on a cluster. Had to pivot to Unity Catalog's Foreign Catalog / Hive Metastore Federation feature instead.

- Snowflake's Glue catalog integration is READ-ONLY. Any attempt to INSERT, UPDATE, or DELETE fails immediately with a clear SQL compilation error. Confirmed via both hands-on testing and Snowflake/AWS documentation — writes to a Glue-owned table must happen via another engine.
- Databricks' Foreign Iceberg / Delta UniForm Ingress is also READ-ONLY for DML. Beyond that, it has a more severe limitation: it cannot READ a Glue Iceberg table at all once that table contains Iceberg V2 delete files (produced by any merge-on-read UPDATE or DELETE, which is Iceberg's default write mode). A single UPDATE performed via Athena rendered the entire table unreadable from Databricks with a hard Scala assertion failure (DeleteFileWrapper), not a graceful degradation. Recovery requires compacting the table (rewriting MoR delete files into CoW data files) from a compatible engine outside Databricks — this was not performed in this POV, so Databricks' read access remained broken for the remainder of testing after Section 7.2.
- Athena required a one-time, easy-to-miss setup step: a query result output S3 location, separate from the Iceberg warehouse path itself.